

PATTERNS IN THE SKY

Astronomers worldwide use not only the same set of 88 constellations, but also the same ways of locating and identifying objects within them.

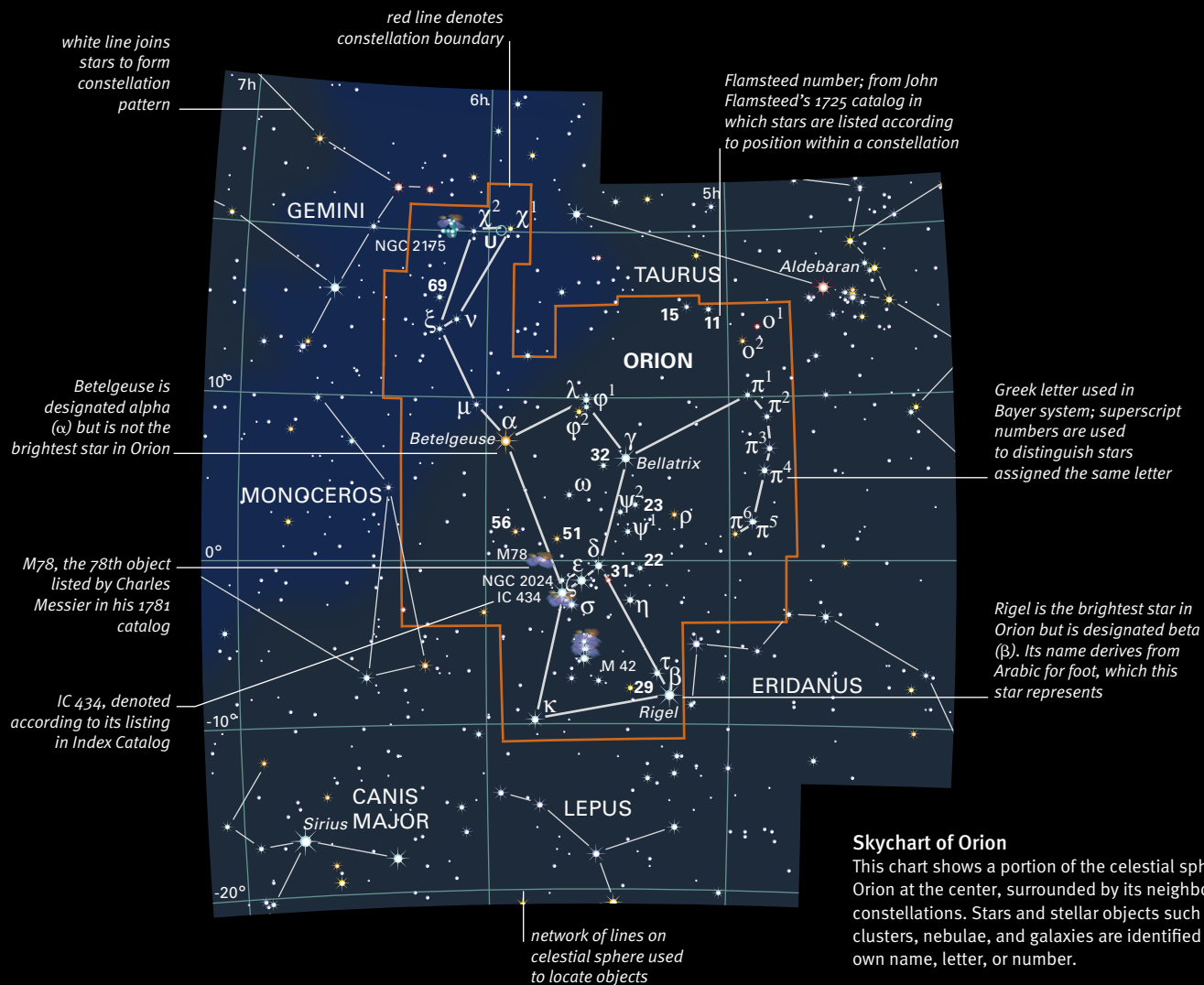
Naming and numbering

Individual stars may be identified by names, numbers, or letters. The brightest stars in a constellation have a Greek letter: alpha (α) is usually the brightest; the next brightest, beta (β); and so on. Once the Greek alphabet is used up, small Roman letters (a, b, c, etc.) are allocated.

Many of the bright stars have names, a large number of which are Arabic in origin. The vast majority, however, are unnamed. Fainter stars, although often within naked-eye visibility, have a number allocated according to their position within the constellation. Stellar objects such as star clusters, nebulae, and galaxies have a catalog number from one of the many volumes that list such objects. NGC numbers, for example, come from the New General Catalog.

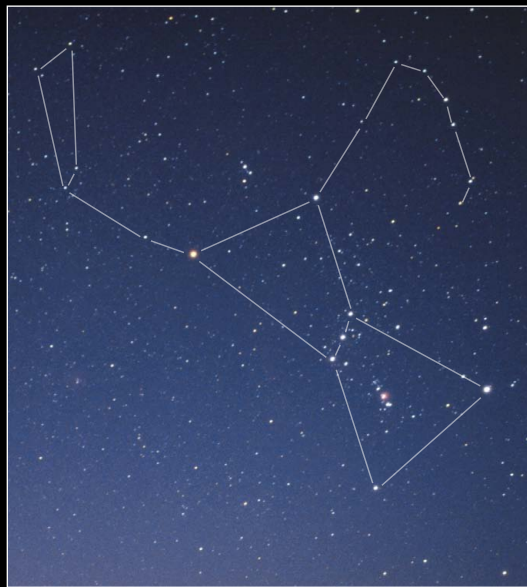
Asterisms

Some stars make a distinctive pattern in the sky but are not a constellation; they are a separate pattern within a constellation, or are made up of stars from more than one constellation. Such a pattern is known as an asterism. The Big Dipper, possibly the best-known pattern in the northern-hemisphere sky, is a prime example. Four stars that form the lower torso of Hercules are known as the Keystone; and the head, neck, and shoulders of Leo is known as the Sickle.



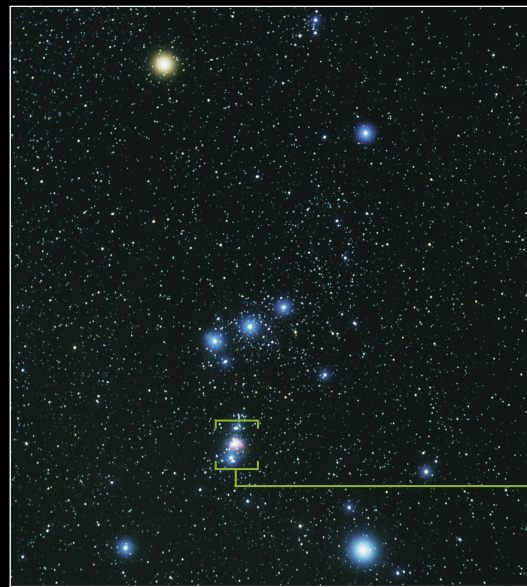
Skychart of Orion

This chart shows a portion of the celestial sphere with Orion at the center, surrounded by its neighboring constellations. Stars and stellar objects such as clusters, nebulae, and galaxies are identified by their own name, letter, or number.



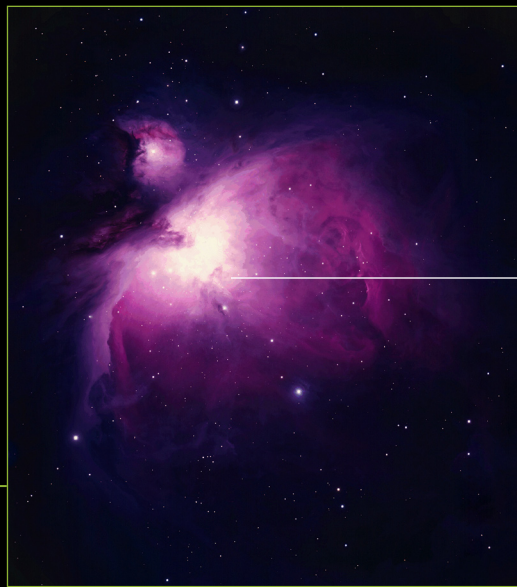
Orion in the night sky

One of the most easily seen constellations in the night sky, Orion is an excellent target for novice star-watchers. Its simple, distinctive shape traces the figure of the mythical hunter. It is seen in both the northern- and southern-hemisphere skies.



Orion's belt and sword

One of Orion's shoulders is marked by the red supergiant Betelgeuse (top left), while blue supergiant Rigel (lower right) is the constellation's brightest star. Three stars form the belt, from which hangs a sword of stars and glowing gas and dust.



at the center of the Orion Nebula is a small cluster of stars called the Trapezium; a small telescope shows it as four stars, while a large one reveals six

Orion Nebula

The Orion Nebula, or M42, is the brightest nebula in the night sky. It is an enormous star-forming cloud of gas and dust some 30 light-years across and 1,500 light-years away. It is illuminated by newly born stars deep within.